
Prompted LLMs Are Conservative Speaker Relabelers: A Transcript-Preserving Pilot on AMI and Simsamu

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Abstract

Speaker-attributed transcripts are only useful when they preserve who said what, but multi-person meetings make speaker labels fragile under overlap, short turns, and noisy diarization. Prior work shows that language models can help speaker attribution when they are fine-tuned or paired with strong acoustic evidence. We ask a narrower practical question: can a lightweight API-based LLM correction layer improve noisy speaker labels while preserving transcript words? We build a controlled pilot from local AMI-SDM and SIMSAMU audio. GPT-4O-MINI-TRANSCRIBE transcribes 90 annotated speech segments, seeded corruption produces noisy speaker labels, and GPT-4.1-MINI relabels the same transcript under no-profile and profile-conditioned prompts. Across 30 window/noise cases, no-profile LLM relabeling reduces mean WDER from 0.390 to 0.375, a 3.8% relative reduction, but the effect is not statistically significant after Holm correction (Wilcoxon one-sided $p = 0.137$, Holm $p = 0.410$). Profile conditioning is weaker, reducing mean WDER to 0.381. The LLM responses are structurally reliable, with 60/60 valid JSON outputs, but they are conservative: no-profile relabeling leaves labels unchanged in 26/30 cases. The improvement appears only in the two-speaker SIMSAMU windows; four-speaker AMI-SDM windows show no LLM gain. These results support LLMs as safe transcript-preserving relabelers, but not as a standalone fix for many-speaker meeting identification.

1 Introduction

Meeting transcripts fail quietly when speaker labels are wrong. The words may look fluent, but a downstream reader, reviewer, or clinical analyst still needs to know who made each claim, asked each question, or accepted each decision. This speaker-attribution problem becomes harder as conversations include more people, more overlap, shorter turns, and less stable timestamps.

Large language models (LLMs) are a natural candidate for correcting these errors because speaker identity is not only acoustic. People reveal roles, preferences, questions, and discourse patterns through text. Prior work has shown that language context can improve diarization and speaker-attributed ASR, especially when LLMs are fine-tuned for transcript-preserving speaker transfer or combined with acoustic speaker probabilities Wang et al. [2024], Park et al. [2023], Li et al. [2026]. The open practical question is smaller but important: can a lightweight API-based LLM relabeler help when we have an ordinary noisy transcript and no custom speech-LLM training?

What does the LLM know without audio? We examine this question in a controlled pilot. We build five evaluation windows from local AMI-SDM and SIMSAMU audio, transcribe real segments with GPT-4O-MINI-TRANSCRIBE, inject seeded speaker-label noise, and ask GPT-4.1-MINI to output speaker IDs only. The relabeler never edits transcript text. We compare raw noisy labels, a deterministic continuity smoother, zero-profile LLM relabeling, and profile-conditioned LLM relabeling. Figure 1 summarizes the pipeline.

Our main result is cautionary. Across 30 window/noise cases, zero-profile LLM relabeling reduces mean WDER from 0.390 to 0.375, a 3.8% relative reduction. This gain is not statistically significant after Holm correction (Wilcoxon one-sided $p = 0.137$, Holm $p = 0.410$). The LLM parses reliably, with 60/60 valid JSON outputs, but it changes few labels: no-profile relabeling leaves 26/30 cases unchanged, improves 3, and worsens 1. The four-speaker AMI-SDM condition shows no LLM improvement at all. The improvement comes from the two-speaker SIMSAMU windows, where no-profile relabeling reduces mean WDER from 0.310 to 0.273.

In summary, our main contributions are:

- We construct a reproducible transcript-preserving pilot benchmark from real local AMI-SDM and SIMSAMU audio, rather than invented transcripts.
- We compare noisy labels, continuity smoothing, zero-profile LLM relabeling, and profile-conditioned LLM relabeling under seeded diarization-style noise.
- We show that prompted LLM relabeling is structurally reliable but conservative: it improves mean WDER by 0.015 overall, yet does not significantly outperform noisy labels.
- We identify a clear many-speaker failure mode: both LLM conditions exactly match noisy-label WDER in four-speaker AMI-SDM windows.

The rest of the paper is organized as follows. Section 2 reviews LLM-based diarization and speaker-attributed ASR. Section 3 describes the benchmark, relabeling prompts, baselines, and metrics. Section 4 reports quantitative results and statistical tests. Section 5 interprets the failure modes and limitations, and section 6 concludes with next steps.

2 Related Work

LLM post-processing for diarization. The closest prior work is Wang et al. [2024], who propose DiarizationLM for speaker diarization post-processing. Their method represents ASR and diarization outputs as speaker-tagged text, uses an LLM to correct speaker assignments, and applies transcript-preserving speaker transfer so the final transcript keeps the original words. Their results show large WDER reductions after fine-tuning, but also warn that zero-shot or lightly prompted models can delete, hallucinate, or fail to improve transcripts. We follow the transcript-preserving principle, but deliberately test the lightweight case: no fine-tuning, no word edits, and API-based speaker-ID-only outputs.

Language-assisted diarization. Park et al. [2023] combine acoustic diarization evidence with lexical probabilities from language models inside contextual beam search, reporting strong relative improvements on speaker-attributed WER. India et al. [2025] similarly show that linguistic cues can help speaker diarization in structured telephone interviews. These approaches support our hypothesis that text contains speaker-attribution information. They also motivate our negative result: lexical context alone is weaker when the candidate speakers are anonymized and the window is short.

Speech-LLM systems for multi-speaker ASR. Several recent systems move beyond text-only correction by conditioning LLMs on audio, diarization, speaker embeddings, or time segments. Serialized-output training with LLM decoders has improved multi-talker ASR on LibriMix and AMI-SDM Shi et al. [2024]. Diarization-aware multi-speaker ASR feeds speaker/time triplets to an LLM backend Lin et al. [2025], while DM-ASR formulates diarization-guided transcription as multi-turn dialogue generation Li et al. [2026]. SpeakerLM and JEDIS-LLM further explore end-to-end or streamable joint ASR and diarization with speaker registration and prompt-cache mechanisms Yin et al. [2025], Shi et al. [2025]. Unlike these systems, our pilot does not train a speech model or expose acoustic embeddings to the LLM. We isolate what a prompted text LLM can do as a low-cost correction layer.

Evaluation and toolkits. Meeting transcription commonly reports word-level speaker-attribution metrics such as WDER, cpWER, and tcpWER. MeetEval standardizes these metrics for meeting transcription systems von Neumann et al. [2023]. pyannote.audio provides widely used neural diarization building blocks and speaker diarization pipelines Bredin et al. [2020], Bredin [2023]. Our experiment uses local diarization annotations as ground truth and computes a small-speaker exact-permutation cpWER-style score locally because the resource-gathering phase could not install MeetEval under the available compiler constraints. This makes our cpWER values useful for within-pilot comparisons, not a substitute for a full MeetEval benchmark.

Dataset	Recording	Segments	Speakers	Mean dur.	Overlap rate	Role in pilot
AMI-SDM	EN2002b.Array1-01	490	4	3.97s	0.714	Meeting window
AMI-SDM	TS3003a.Array1-01	242	4	4.24s	0.479	Meeting window
AMI-SDM	ES2004c.Array1-01	497	4	4.52s	0.622	Meeting window
SIMSAMU	dj_mai_malaise_stef	80	2	3.11s	0.050	Clinical dialogue
SIMSAMU	dj_2022_douleur_thoracique	79	2	2.69s	0.000	Clinical dialogue

Table 1: Datasets and selected recordings. AMI-SDM provides four-speaker meeting windows with high overlap; SIMSAMU provides two-speaker dialogue windows used as a sanity condition.

Datasets. AMI-SDM is a standard meeting benchmark derived from the AMI Meeting Corpus Carletta et al. [2006]. It is important for this study because it has four-speaker meeting-style interactions and high overlap rates in the selected shard. SIMSAMU provides a smaller two-speaker clinical-dialogue sanity dataset Medkit [2023]. Using both lets us compare the easier two-speaker setting against the many-speaker condition that motivated the study.

3 Methodology

Task. We study transcript-preserving speaker relabeling. Each evaluated segment i has transcript text x_i , true speaker y_i , noisy speaker label $\hat{y}_i$, and word count n_i . A correction method predicts $\hat{y}_i$ while keeping x_i fixed. The goal is to reduce speaker-attribution error without changing, deleting, or inserting words.

Data construction. We use local AMI-SDM and SIMSAMU resources. AMI-SDM contributes three four-speaker meeting windows from English single distant microphone audio; SIMSAMU contributes two two-speaker French clinical-dialogue windows. The local AMI-SDM shard contains speaker time segments but not manual word transcripts, so we avoid invented text by transcribing selected real audio chunks with GPT-4O-MINI-TRANSCRIBE. Ground-truth speaker IDs come from local diarization annotations. Each evaluation window uses clean unscored turns for optional speaker profiles and scored turns for metric computation.

Noise model. We simulate diarization-style speaker-label errors by corrupting the true speaker label of each scored turn. Corruption is seeded and uses low, medium, and high base-noise settings. The corruption probability increases for overlapping turns, short turns, and speaker-change boundaries, matching common diarization failure surfaces. We use experiment seed 42 for setup and noise seeds 101 and 202 for each noise level.

Methods. We compare four conditions.

- NOISY LABELS keeps the seeded corrupted speaker labels and serves as the direct baseline.
- CONTINUITY SMOOTHING fixes isolated single-turn speaker changes when the neighboring turns agree.
- LLM NO-PROFILE gives GPT-4.1-MINI the noisy transcript context and candidate speaker IDs, then asks for JSON speaker assignments only.
- LLM PROFILE uses the same relabeler but adds clean, unscored high-confidence profile turns for each speaker.

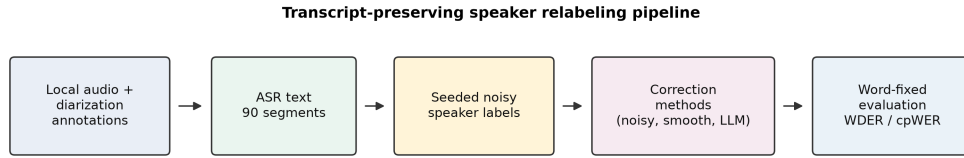
The LLM uses temperature 0 where accepted, JSON-object response format, and a speaker-ID-only output schema. Fallback models were implemented but not used in the run.

Metrics. Our primary metric is word diarization error rate:

$$\text{WDER} = \frac{\sum_i n_i \mathbb{I}[\hat{y}_i \neq y_i]}{\sum_i n_i}. \quad (1)$$

We also report a local cpWER-style score that concatenates words by predicted and true speaker, then chooses the minimum-error speaker permutation for the small number of speakers in each window. Segment error is the fraction of scored segments with the wrong speaker label. Speaker count MAE is the absolute error in the number of speakers used by a method. Parse success is the fraction of LLM responses that return valid JSON assignments with usable segment IDs.

Statistical testing. We compare each correction method against NOISY LABELS across 30 paired case pairs. Because normality was not supported for WDER differences, we use one-sided Wilcoxon



The LLM can only return speaker IDs; transcript words remain fixed for all methods.

Figure 1: Overview of the transcript-preserving pilot. We transcribe real audio segments, inject seeded speaker-label noise, run deterministic and LLM correction methods, and evaluate speaker labels while keeping transcript words fixed.

Method	Mean WDER ↓	Mean cpWER ↓	Segment err. ↓	Speaker-count MAE ↓	Parse success
LLM NO-PROFILE	0.375	0.521	0.373	0.400	1.00
LLM PROFILE	0.381	0.529	0.380	0.400	1.00
NOISY LABELS	0.390	0.550	0.387	0.400	1.00
CONTINUITY SMOOTHING	0.442	0.628	0.437	0.400	1.00

Table 2: Overall results across 30 window/noise cases. Lower is better for WDER, cpWER, segment error, and speaker-count MAE. Best values are in **bold**. LLM NO-PROFILE gives the best mean accuracy, but the margin over NOISY LABELS is small.

signed-rank tests, bootstrap 95% confidence intervals for mean WDER reduction, paired effect size d_z , and Holm correction across the three main comparisons.

4 Results

Overall performance. Table 2 shows the aggregate results across 30 window/noise cases. LLM NO-PROFILE achieves the lowest mean WDER (0.375) and mean CPWER (0.521), improving over NOISY LABELS by 0.015 WDER and 0.030 cpWER. LLM PROFILE is slightly weaker, with mean WDER 0.381 and mean CPWER 0.529. CONTINUITY SMOOTHING performs worst on WDER and segment error, showing that simple local continuity is not a reliable substitute for speaker-aware reasoning.

Dataset split. The overall mean hides a clear split by speaker count and dataset. In AMI-SDM, both LLM conditions exactly match NOISY LABELS on mean WDER (0.443) and mean CPWER (0.630). In SIMSAMU, LLM NO-PROFILE reduces mean WDER from 0.310 to 0.273 and mean CPWER from 0.431 to 0.357. This supports the narrower claim that prompted LLMs can correct some two-speaker transcript-context errors, but weakens the broader hypothesis for four-speaker meeting identification.

Statistical evidence. Table 4 reports paired tests against NOISY LABELS. No correction method achieves a statistically significant WDER improvement after Holm correction. LLM NO-PROFILE has a positive mean WDER reduction of 0.015, but its 95% bootstrap interval includes zero ($[-0.009, 0.045]$), and the Holm-adjusted p -value is 0.410. LLM PROFILE shows an even smaller mean reduction of 0.009. CONTINUITY SMOOTHING has a negative reduction, meaning it increases mean WDER relative to NOISY LABELS.

Conservatism and parse reliability. The LLM did not fail structurally: all 60 responses parsed successfully. The main failure mode is conservatism. LLM NO-PROFILE leaves 26/30 cases unchanged, improves 3, and worsens 1. LLM PROFILE leaves 28/30 cases unchanged, improves 1, and worsens 1. Figure 2 and figure 3 show the same pattern visually: the average LLM gain is small, concentrated in SIMSAMU, and absent in AMI-SDM.

Dataset	Method	Speakers	Mean WDER ↓	Mean cpWER ↓
AMI-SDM	NOISY LABELS	4	0.443	0.630
AMI-SDM	LLM NO-PROFILE	4	0.443	0.630
AMI-SDM	LLM PROFILE	4	0.443	0.630
AMI-SDM	CONTINUITY SMOOTHING	4	0.453	0.626
SIMSAMU	NOISY LABELS	2	0.310	0.431
SIMSAMU	LLM NO-PROFILE	2	0.273	0.357
SIMSAMU	LLM PROFILE	2	0.287	0.379
SIMSAMU	CONTINUITY SMOOTHING	2	0.425	0.631

Table 3: Dataset split. The LLM gain is concentrated in the two-speaker SIMSAMU condition. In four-speaker AMI-SDM, both LLM conditions leave aggregate WDER and cpWER unchanged relative to NOISY LABELS.

Comparison	Mean WDER red.	95% CI	Test	<i>p</i>	Holm <i>p</i>
NOISY LABELS vs. CONTINUITY SMOOTHING	-0.052	$[-0.101, -0.008]$	Wilcoxon	0.978	0.978
NOISY LABELS vs. LLM NO-PROFILE	0.015	$[-0.009, 0.045]$	Wilcoxon	0.137	0.410
NOISY LABELS vs. LLM PROFILE	0.009	$[-0.001, 0.029]$	Wilcoxon	0.327	0.655

Table 4: Paired WDER tests over 30 case pairs. Positive reduction means the method improves over NOISY LABELS. No correction method reaches statistical significance after Holm correction.

5 Discussion

What worked? The strongest positive result is not accuracy, but controllability. A simple JSON speaker-ID prompt produced valid outputs in all 60 LLM calls and preserved transcript text by construction. This matters because prior LLM post-processing can degrade transcripts when the model is allowed to rewrite words Wang et al. [2024]. In this pilot, the LLM correction layer is safe in the narrow sense that it does not alter ASR text.

What failed? The LLM rarely changed labels in four-speaker AMI-SDM windows. Many local speaker IDs are anonymized, and short meeting snippets do not provide stable lexical identities. Without acoustic confidence, embeddings, names, roles, or longer history, a text-only LLM has little evidence for deciding that speaker S1 should become speaker S3. This explains why both LLM conditions exactly match NOISY LABELS on AMI-SDM. The model’s conservatism is preferable to hallucinated relabeling, but it limits correction value.

Why did profiles not help? We expected profile-conditioned relabeling to outperform no-profile relabeling because profile turns could anchor each speaker. The result does not support that hypothesis. The profiles were short, text-only, and unscored; they did not provide robust identity evidence in meeting snippets. In SIMSAMU, the profile condition improves over NOISY LABELS but underperforms LLM NO-PROFILE. This suggests that weak profiles can add prompt complexity without adding enough speaker evidence.

Why did continuity smoothing hurt? CONTINUITY SMOOTHING assumes that isolated speaker changes are likely errors. That assumption fails in dialogue where true turns alternate quickly. The method increases mean WDER from 0.390 to 0.442, and it is especially poor in SIMSAMU low-noise cases. This failure is useful: a transcript correction layer needs discourse-aware evidence, not only local temporal smoothness.

Limitations. This is a pilot, not a benchmark claim. The experiment has five windows and 30 noisy cases. The AMI-SDM shard lacks manual word transcripts, so ASR text is a pseudo-reference for words and the evaluation isolates speaker-label correction rather than ASR word accuracy. Noise injection is controlled and seeded, but it simulates diarization errors rather than using a full WhisperX plus pyannote cascade. The speaker profiles are short and text-only. Finally, our cpWER-style score is a local exact-permutation implementation for small speaker counts, not an official MeetEval run.

Broader implication. The result points to a practical design rule: use prompted LLMs as constrained, transcript-preserving relabelers only when they receive stronger speaker evidence. Acoustic confidence scores, speaker embeddings, enrollment profiles, longer conversational history, or

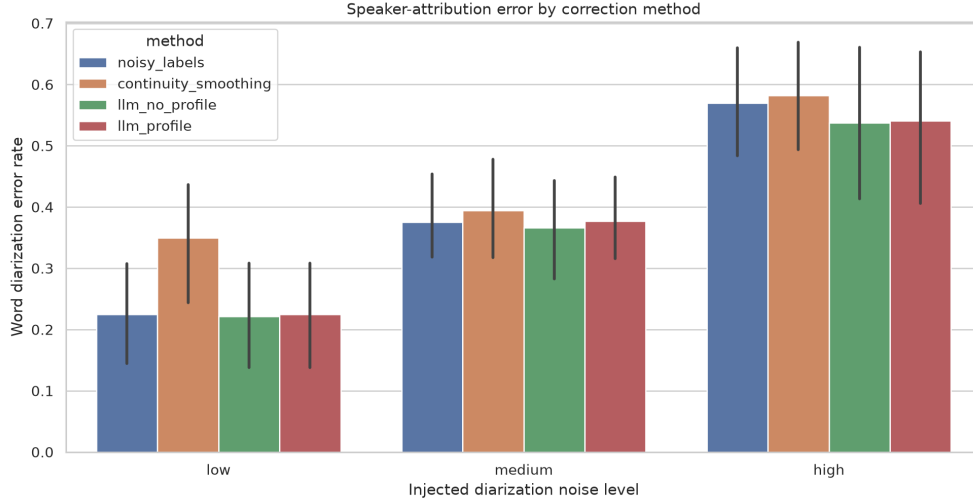


Figure 2: WDER by method across all 30 cases. Lower is better. LLM NO-PROFILE has the best mean WDER, but the improvement over NOISY LABELS is small and not statistically significant.

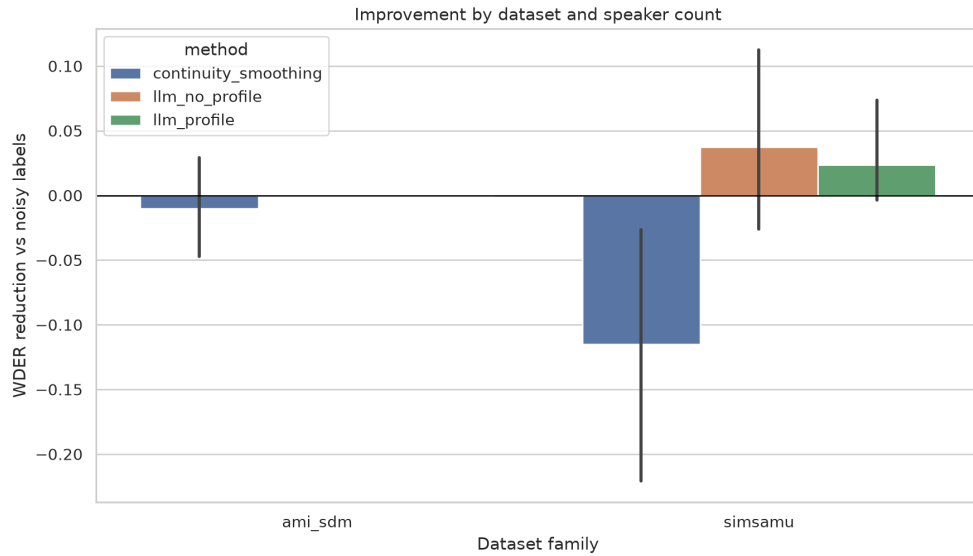


Figure 3: WDER reduction relative to NOISY LABELS by dataset. Positive values mean improvement. LLM gains appear in the two-speaker SIMSAMU setting, while the four-speaker AMI-SDM setting shows no LLM reduction.

fine-tuned DiarizationLM-style models are likely necessary for robust many-speaker meeting identification.

6 Conclusion

We tested whether a real API-based LLM can improve multi-person speaker identification from noisy ASR-style transcripts while preserving every transcript word. In a controlled pilot over local AMI-SDM and SIMSAMU audio, no-profile LLM relabeling reduced mean WDER from 0.390 to 0.375, but the improvement was not statistically significant after Holm correction. The LLM was reliable as a structured output system, with 100% parse success, but conservative as a speaker corrector.

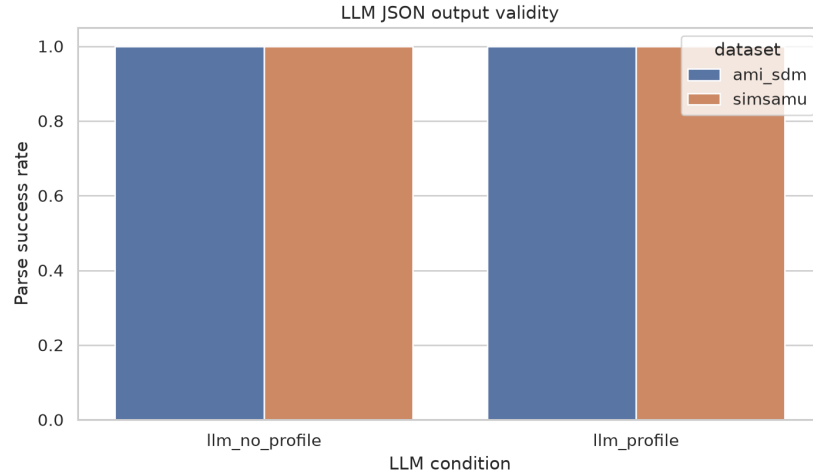


Figure 4: LLM parse success. Both LLM conditions return valid JSON assignments for every call, so the limiting factor is not output formatting but whether the model changes incorrect labels.

The key finding is that prompted text-only LLM relabeling helps only weakly and does not fix four-speaker meeting attribution in this setting. Future work should run the same harness on full AMI and ICSI with manual transcripts, replace simulated noise with real WhisperX plus pyannote outputs, add acoustic confidence and speaker-embedding summaries to the prompt, and compare zero-shot API prompting against fine-tuned transcript-preserving models.

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A Reproducibility Details

The experiment used GPT-4O-MINI-TRANSCRIBE for 90 audio segments totaling 343.28 seconds and GPT-4.1-MINI for 60 LLM calls. The LLM calls consumed 63,822 input tokens and 19,800 output tokens. Using the OpenAI prices recorded in the run metadata on July 6, 2026, the estimated API cost was \$0.017 for ASR, \$0.057 for LLM calls, and \$0.074 total OpenAI [2026a,b].

The local environment recorded Python 3.12.8, NumPy 2.2.6, SciPy 1.18.0, scikit-learn 1.9.0, statsmodels 0.14.6, OpenAI 2.44.0, Matplotlib 3.11.0, seaborn 0.13.2, datasets 5.0.0, and pyarrow 24.0.0. GPU acceleration was not required; the environment contained four NVIDIA RTX A6000 GPUs with 49 GB each.

The primary command sequence is:

```
source .venv/bin/activate
uv add openai matplotlib scikit-learn statsmodels seaborn
python src/run_experiment.py
```

The main generated files are `results/metrics_by_case.csv`, `results/aggregate_metrics.csv`, `results/statistical_tests.csv`, `results/error_analysis.csv`, `results/usage_summary.json`, cached ASR outputs under `results/cache/`, cached LLM outputs under `results/model_outputs/`, and generated figures under `figures/`. Re-running with existing caches should reproduce the reported metrics without new API calls.